

week-5-lesson-1

example-1

Print 200 with and without a sign.

example-2

Print unsigned integer 40000 left justified in a 15-digit field with 8 digits.

example-3

Read a time of the form hh:mm:ss, storing the parts of the time in the integer variables hour, minute and second. Skip the colons (:) in the input stream. Use the assignment suppression character.

example-4

Read a time of the form hh:mm:ss, storing the parts of the time in the integer variables hour, minute and second. Skip the colons (:) in the input stream. Do not use the assignment suppression character.

example-5

Write a program to test the difference between the %d and %i conversion specifiers when used in scanf statements. Use the statements

```
scanf( "%i%d", &x, &y );
```

```
printf( "%d %d\n", x, y );
```

to input and print the values. Test the program with the following sets of input data:

10 10

-10 -10

010 010

0x10 0x10

example-6

Write a program to access a file and implement following statements:

Write a statement that opens the file "old_data.txt" for reading and assigns the returned file pointer to ofPtr.

Write a statement that opens the file "transactions.txt" for reading and assigns the returned file pointer to tfPtr.

Write a statement that opens the file "new_data.txt" for writing (and creation) and assigns the returned file pointer to nfPtr.

Write a statement that reads a record from the file "old_data.txt". The record consists of integer accountNumber, string name and floating-point currentBalance.

Write a statement that reads a record from the file "transactions.txt". The record consists of the integer accountNumber and floating-point dollarAmount.

Write a statement that writes a record to the file "new_data.txt". The record consists of the integer accountNumber, string name and floating-point currentBalance.

Use the account number on each file as the record key for matching purposes. Assume that each file is a sequential file with records stored in increasing account number order.

When a match occurs (i.e., records with the same account number appear on both the old_data file and the transactions file), add the dollar amount on the transaction file to the current balance on the master file and write the "new_data.txt" record. (Assume that purchases are indicated by positive amounts on the transaction file, and that payments are indicated by negative amounts.) When there is an only record for a particular account in old_data.txt or transactions.txt but no corresponding transaction record or old_data, merely write the new record to "new_data.txt".

example-7

Write a C program that reads a character stream from standard input one character at a time using getchar() until reaching an end-of-file condition (EOF).

Task Requirements:

- Read characters inside a while loop using `(c = getchar()) != EOF`.
- Immediately echo each character back to standard output using `putchar(c)`.
- Maintain a counter of all non-newline characters read from the stream.
- After the loop terminates (on EOF), print the final character count using `printf()`.

Expected Output Example:

Input: Hello World!

Echo Output: Hello World!

Total non-newline characters read: 12